EE1100: Basic Electrical Engineering
Tutorial 7: Transformer

Q1. The emf per turn for a single phase 2310/220 V, 50 Hz transformer is approximately 13 volts. Calculate (a) the number of primary and secondary turns and (b) the net cross-sectional area of the core, for a maximum flux density of $1.4\,Wb/m^2$.

Ans.: (a) 189, 18 (b) 393 cm^2

Q2. (a) A 2200/220 V 50 Hz, single phase transformer has exciting current of 0.6 A and a core loss of 361 watts, when its high voltage side is energised at rated voltage. Calculate the two components of the exciting current.

(b) If the transformer of part (a), supplies a load current of 60 A at 0.8 p.f. lag on its low voltage side, then calculate the primary current and its power factor. Ignore leakage impedance drops.

Ans.: (a) Ic=0.164 A, I_Φ=0.577 A (b) 6.48 A, 0.766 (lag)

Q3. A single-phase transformer is designed to operate at 240/120 V, 50 Hz. Calculate the secondary no load voltage and its frequency if the high voltage side of the transformer is connected to:
(a) 240 V, 40 Hz (b) 120 V, 25 Hz (c) 120 V, 50 Hz (d) 480 V, 50 Hz (e) 240 V, dc.

Ans.: (a) 120 V, 40 Hz (b) 60 V, 25 Hz (c) 60 V, 50 Hz (d) Overrated, 240 V, 50 Hz
(e) Transformer may burn

Q4. A 33 kVA, 2200/220 V, 50 Hz single phase transformer has the following parameters:
Primary winding (*h.v.* side): resistance $r_1 = 2.4\,\Omega$; leakage reactance $x_1 = 6.00\,\Omega$
Secondary winding (*l.v.* side): resistance $r_2 = 0.03\,\Omega$; leakage reactance $x_2 = 0.07\,\Omega$

(a) Find the primary resistance and leakage reactance referred to secondary.
(b) Find the secondary resistance and leakage reactance referred to primary.
(c) Find the equivalent resistance and equivalent leakage reactance referred to (i) primary (ii) secondary.
(d) Calculate total ohmic loss at full load.
(e) Calculate the voltage to be applied to the high voltage side, in order to obtain a short circuit current of l60 A in the low voltage winding. Under these conditions, find the power input also.

Ans.: (a) 0.024, 0.06 Ω (b) 3, 7 Ω, (c) (i) 5.4, 13 Ω (ii) 0.054, 0.13 Ω (d) 1216 W (e) 225 V, 1382 W

Q5. A 10 kVA, 2500/250 V, single phase transformer has resistances and leakage reactance as follows:

$r_1 = 4.8\,\Omega$; $r_2 = 0.048\,\Omega$;
$x_1 = 11.2\,\Omega$; $x_2 = 0.112\,\Omega$;

Subscripts 1 and 2 denote high voltage and low voltage windings respectively. With primary supply voltage held constant at 2500 V, calculate the secondary terminal voltage, when
(a) the low voltage (LV) winding is connected to a load impedance of $5 + j3.5\,\Omega$.
(b) the transformer delivers its rated current at 0.8 p.f. lagging, on the LV side.

Ans.: (a) 242 V (b) 241.6 V

Q6. The equivalent circuit referred to the low-tension side of a 2500/250 V single phase transformer is shown in Fig. 1. The load impedance referred to the high-tension terminals is 380+ j230 Ω. Compute

(a) The secondary terminal voltage
(b) Primary current and power factor,
(c) Power output and efficiency.

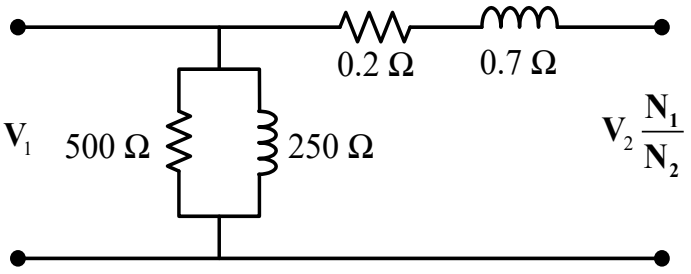


Fig. 1

Ans.: (a) 222 V (b) 5.1 A, 0.794 (lag) (c) 9500 W, 93.82%

Q7. A 20 kVA, 2500/250 V, 50Hz, single-phase transformer gave the following test results:
Open circuit test (in *l.v.* side): 250V, 1.4 A, 105 watts
Short circuit test (in *h.v.* side): 104 V, 8 A, 320 watts.

Compute the parameters of the approximate equivalent circuit referred to high voltage and low voltage sides. Also draw the exact equivalent circuit referred to the low voltage side.

Ans.: HV side: $R_{1H} = R'_{1L} = 2.5$, $X_{1H} = X'_{1L} = 6$, $R_{cH} = 59524$, $X_{\Phi H} = 18726\ \Omega$
LV side: $R_{1L} = R'_{1H} = 0.025$, $X_{1L} = X'_{1H} = 0.06$, $R_{cL} = 595.24$, $X_{\Phi L} = 187.26\ \Omega$

Q8. A 20 kVA, 2500/500 V, 50 Hz, single-phase transformer has the following parameters.

h.v. winding	*l.v.* winding
$r_1 = 8\ \Omega$;	$r_2 = 0.3\ \Omega$;
$x_1 = 17\ \Omega$;	$x_2 = 0.7\ \Omega$;

Find the voltage regulation and the secondary terminal voltage at full load for a p.f. of
(a) 0.8 lagging and (b) 0.8 leading with primary voltage held constant at 2500 V.

Ans.: (a) 10.59%, 447.04 V (b) -2.65%, 513.28 V

Q9. A 10 kVA, 2500/250 V, single-phase transformer gave the following test results.
Open circuit test (in *l.v.* side) – 250V, 0.8 A, 50 W.
Short circuit test (in *h.v.* side) – 60 V, 3 A, 45 W.

(a) Calculate the efficiency at $\frac{1}{4}$, $\frac{1}{2}$, 1 and $1\frac{1}{4}$ of full load at 0.8 p.f.
(b) Calculate the load (kVA output) at which maximum efficiency occurs and also the value of maximum efficiency at 0.8 p.f.
(c) Compute the voltage regulation and the secondary terminal voltage under rated load at power factors of (i) 0.8 lagging and (ii) 0.8 leading.

Ans.: (a) 97.32%, 98.28%, 98.40%, 98.28% (b) 7.905 kVA, 98.44% (c) (i) 2.5%, 243.75, (ii) -1.21%, 253.025 V

Q10. The maximum efficiency of a 100 kVA, single-phase transformer is 98% and occurs at 80% of full load at 0.8 p.f. If the leakage impedance of the transformer is 5%, find the voltage regulation at rated load of 0.8 p.f. lagging.

Ans.: 3.75%